

# Test Automation

Backend

1st edition

## Module 4: CRUD and Playwright

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# CRUD

C → Create

R → Read

U → Update

D → Delete

# Postman

Simple and fast to learn



POSTMAN

# Pro/Con

| PRO               | CON                                |
|-------------------|------------------------------------|
| Easy to start     | No version control                 |
| Fast              | Requires payment to other features |
| Snippets          | Not so escalable                   |
| CI CD integration | Dificult to create complex flows   |

# Playwright

Playwright is an open-source automation framework used for testing web applications and APIs.

It allows developers and QA engineers to write automated tests using languages like JavaScript or TypeScript, simulating real user interactions or making direct API requests. With Playwright, you can create reliable, scalable, and maintainable test suites that run across different browsers and environments.



# Playwright

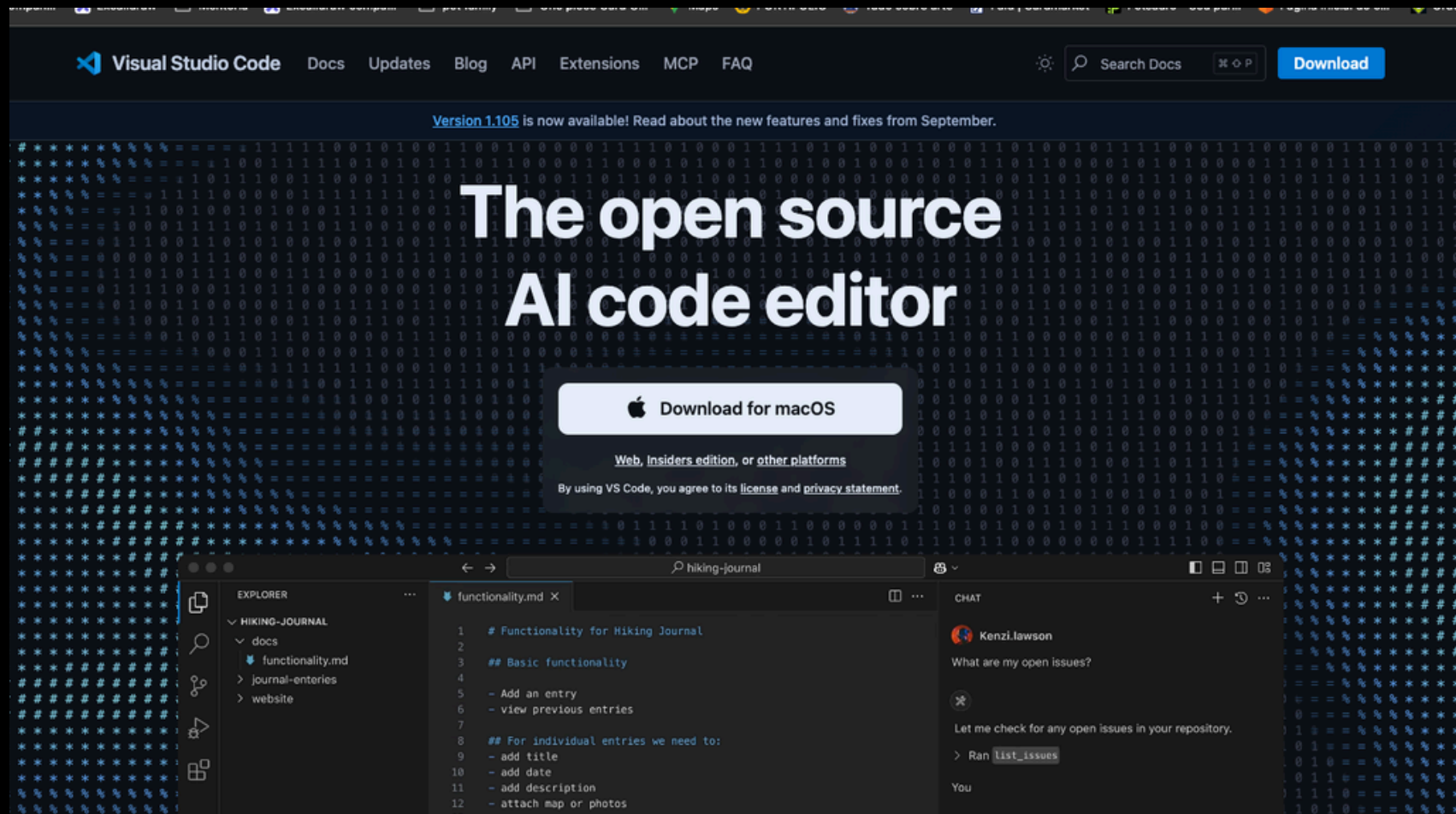


- True cross-browser support
- Fast and reliable
- Unified testing platform — supports UI, API, accessibility, and visual testing natively.
- Multi-tab and multi-domain testing — great for complex applications and real-world scenarios.
- Developer-friendly
- Built-in reports, traces, and videos
- CI/CD ready — simple integration with GitHub Actions, Jenkins, or Azure Pipelines.
- Great debugging tools — --ui mode, visual trace viewer, and network inspection

# Postman vs Playwright

| Aspect            | Postman | Playwright    |
|-------------------|---------|---------------|
| Ease of Use       | High    | Medium        |
| Automation        | Limited | Very Powerful |
| Reusability       | Low     | High          |
| Scalability       | Low     | High          |
| CI/CD Integration | Limited | Excellent     |

# VSCode

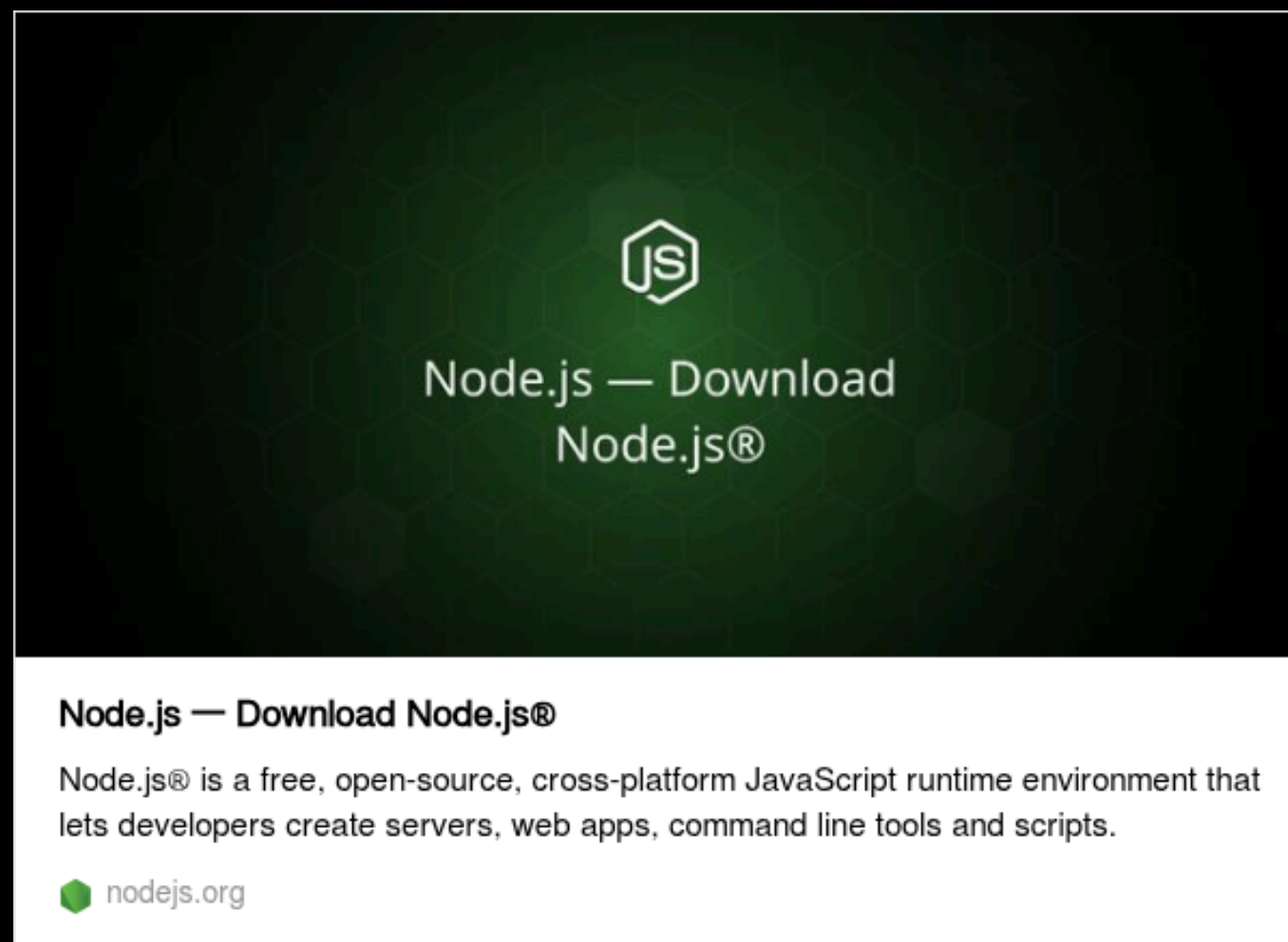


**A lightweight and powerful source-code editor developed by Microsoft.**

**It supports multiple programming languages, has built-in Git integration, and offers extensions for debugging, linting, and automation tools like Playwright.**

**<https://code.visualstudio.com>**

# Node and NPM



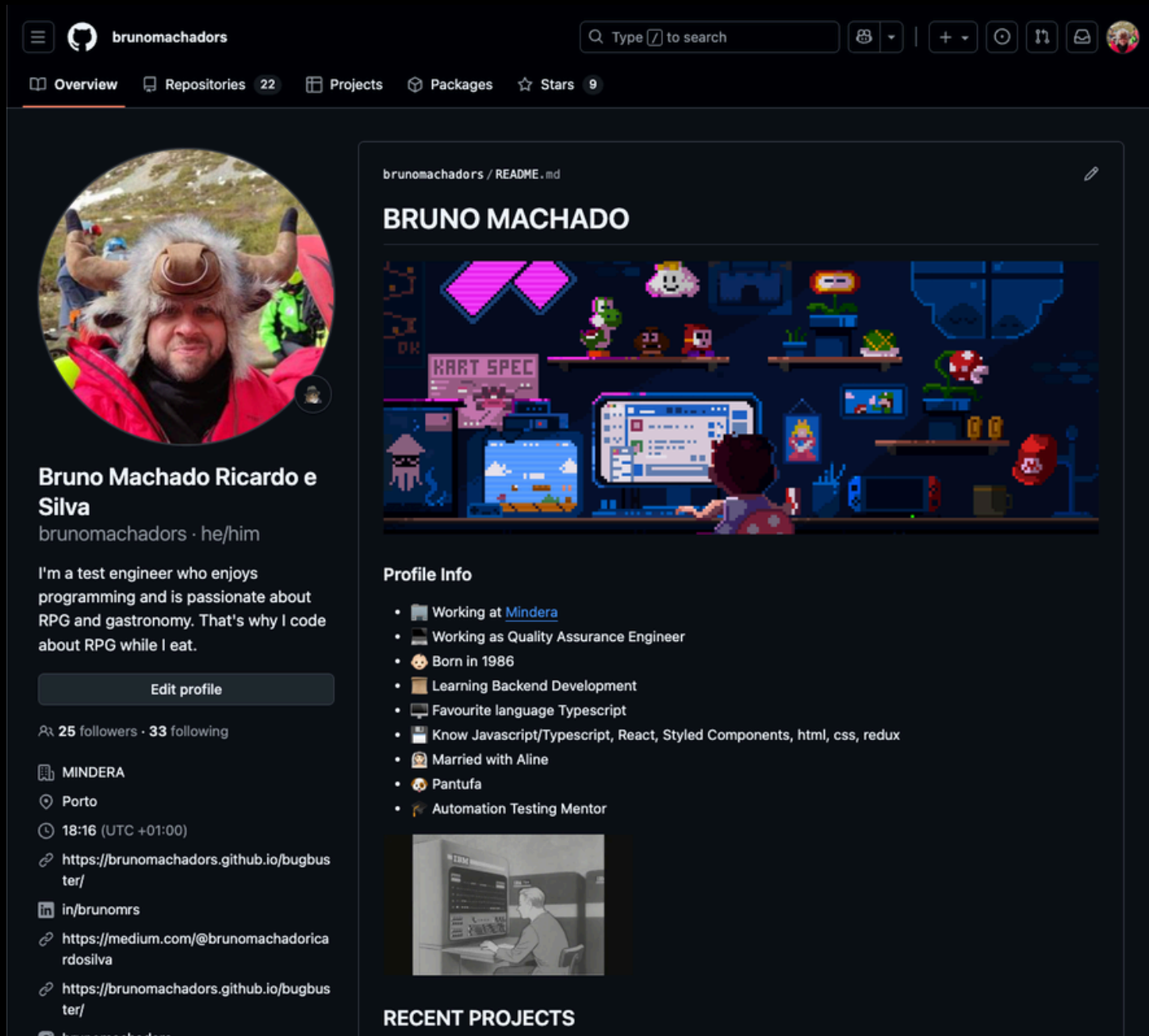
Node.js is like a huge library full of tools that let JavaScript run outside the browser.

npm (Node Package Manager) is the librarian — it helps you find, install, and organize the packages (books) you need from that library.

Together, they make it easy to build and run automation scripts, web apps, and Playwright tests.

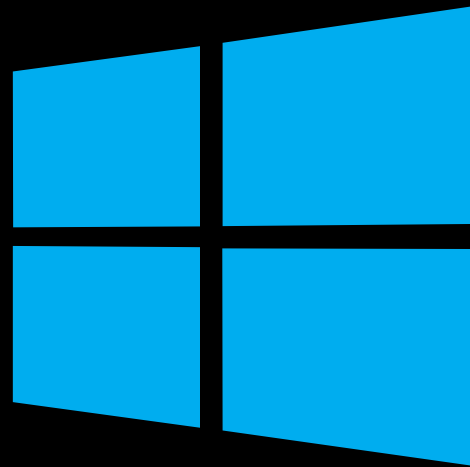
<https://nodejs.org/en/download>

# Git

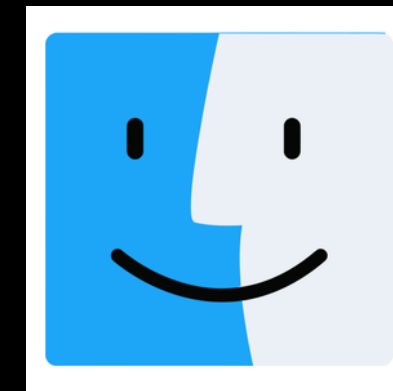


A distributed version control system that tracks changes in your codebase. It allows teams to collaborate efficiently, manage branches, and maintain version history for every change made to the project.

# How to install git



<https://git-scm.com/install/>



<https://brew.sh>

**brew install git**

# How to check git?

**git -v**

```
git version 2.50.1 (Apple Git-155)
```

## How to check git?

```
git config --global user.name
```

```
git config --global user.email
```

```
Bruno Machado  
brunomachadors@gmail.com
```

# How to check git?

```
git config --global user.name "Your New GitHub Name"  
git config --global user.email "your-new-email@example.com"
```

# npm init playwright@latest

```
Need to install the following packages:
```

```
create-playwright@1.17.139
```

```
Ok to proceed? (y) y
```

```
> npx
```

```
> "create-playwright"
```

```
Getting started with writing end-to-end tests with Playwright:
```

```
Initializing project in '.'
```

```
✓ Do you want to use TypeScript or JavaScript? · JavaScript
```

```
✓ Where to put your end-to-end tests? · tests
```

```
✓ Add a GitHub Actions workflow? (Y/n) · true
```

```
✓ Install Playwright browsers (can be done manually via 'npx playwright install')? (Y/n) · true
```

# Recommended commands

```
npx playwright test
```

Runs the end-to-end tests.

```
npx playwright test --ui
```

Starts the interactive UI mode.

```
npx playwright test --project=chromium
```

Runs the tests only on Desktop Chrome.

```
npx playwright test example
```

Runs the tests in a specific file.

```
npx playwright test --debug
```

Runs the tests in debug mode.

```
npx playwright codegen
```

Auto generate tests with Codegen.

# npx playwright test

```
Running 6 tests using 6 workers  
6 passed (11.1s)
```

To open last HTML report run:

```
npx playwright show-report
```

# npx playwright show-report

|                                        |                    |            |          |         |           |      |
|----------------------------------------|--------------------|------------|----------|---------|-----------|------|
| Q                                      | All 6              | ✓ Passed 6 | Failed 0 | Flaky 0 | Skipped 0 | ⚙    |
| 21/10/2025, 17:25:47 Total time: 11.1s |                    |            |          |         |           |      |
| ✓ example.spec.js                      |                    |            |          |         |           |      |
| ✓                                      | has title          | chromium   |          |         |           |      |
|                                        | example.spec.js:4  |            |          |         |           | 4.3s |
| ✓                                      | get started link   | chromium   |          |         |           |      |
|                                        | example.spec.js:11 |            |          |         |           | 3.8s |
| ✓                                      | has title          | firefox    |          |         |           |      |
|                                        | example.spec.js:4  |            |          |         |           | 1.9s |
| ✓                                      | get started link   | firefox    |          |         |           |      |
|                                        | example.spec.js:11 |            |          |         |           | 1.8s |
| ✓                                      | has title          | webkit     |          |         |           |      |
|                                        | example.spec.js:4  |            |          |         |           | 1.2s |
| ✓                                      | get started link   | webkit     |          |         |           |      |
|                                        | example.spec.js:11 |            |          |         |           | 1.2s |

# npx playwright --ui

